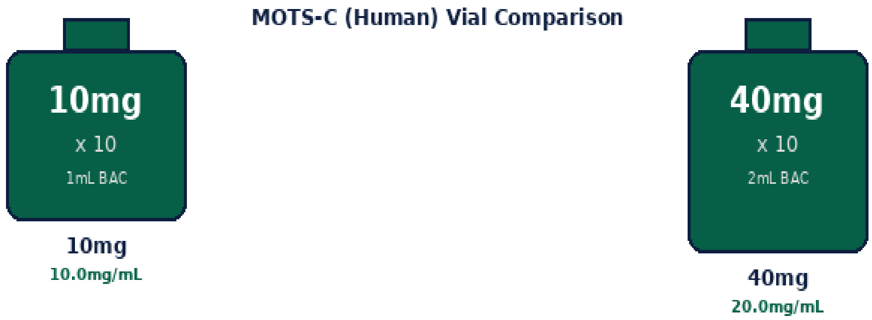
# MOTS-C (Human)

**Mitochondrial-Derived Peptide | 10mg & 40mg x 10 Vials**

AMPK Activation | Metabolic Regulation | Exercise Mimetic

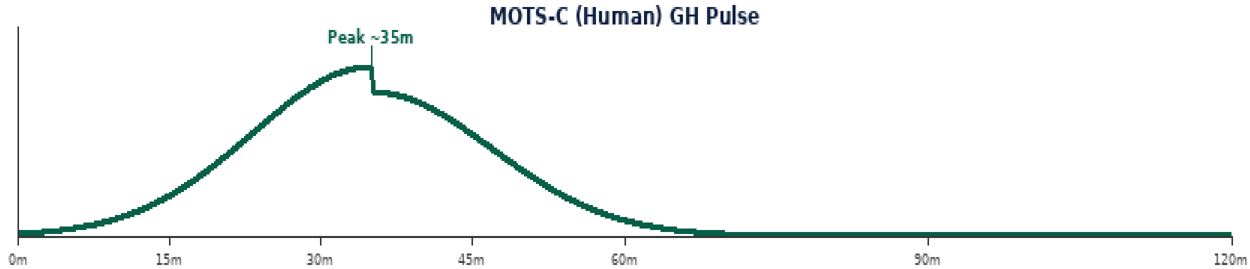
Purity: 99.8% | Endotoxin-Free | Research Use Only

**MOTS-C is a mitochondrial-derived peptide (MDP) encoded by the 12S rRNA gene. It is a powerful AMPK activator and metabolic regulator -- called an exercise mimetic for its ability to activate energy-sensing pathways independently of exercise. Doses can range from 1mg to 5mg per injection.**




| Parameter | Specification |
|---|---|
| **Class** | Mitochondrial-derived peptide (MDP) -- AMPK activator |
| **Vial Sizes** | 10mg x 10 \| 40mg x 10 \| Purity 99.8% |
| **Reconstitution** | 10mg: 1mL BAC->10mg/mL \| 40mg: 2mL BAC->20mg/mL |
| **Standard Dose** | 1,000-5,000 mcg (1-5mg) per injection \| no fasting required |
| **Frequency** | 1x daily or 3-5x per week SubQ \| morning preferred |
| **Cycle & Storage** | 8-12 weeks on, 4 weeks off \| 2-8 deg C \| 28 days reconstituted |

## MECHANISM OF ACTION & RESEARCH BENEFITS

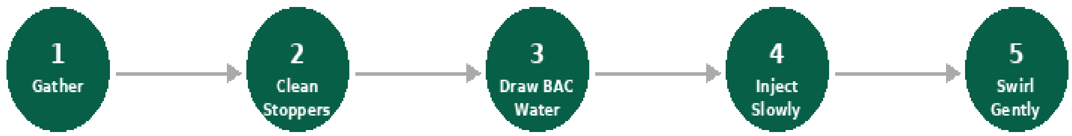



| Pathway | Mechanism | Benefit |
|---|---|---|
| AMPK Activation | Activates AMP-activated protein kinase; cellular energy sensor; inhibits anabolic pathways; activates fat-burning; mimics calorie restriction and exercise signalling | Metabolic improvement; fat oxidation; glucose regulation |
| Folate Cycle | Regulates folate-methionine cycle; affects one-carbon metabolism | Metabolic regulation; longevity research |
| Anti-Aging & Exercise | Endogenous MOTS-C declines with age; supplementation restores youthful metabolic signalling and physical performance in animal models | Anti-aging research; exercise mimetic; longevity |

| Benefit | Context |
|---|---|
| **Exercise Mimetic** | MOTS-C activates AMPK and metabolic pathways of physical exercise without exercise itself. |
| **Insulin Sensitivity** | Improves glucose uptake and insulin sensitivity in multiple research models. |
| **Anti-Aging** | Endogenous MOTS-C declines with age; supplementation restores metabolic function in animal models. |

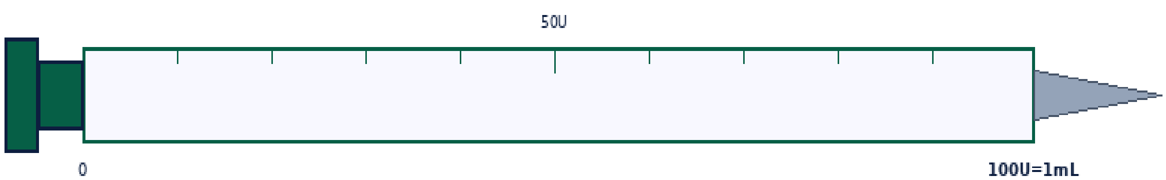
## RECONSTITUTION & DOSE CALCULATION

**All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.**




1 mL BAC Water per vial. Roll gently -- NEVER shake.

| Step | Action | Detail |
|---|---|---|
| 1 | Gather | Vial, BAC Water, insulin syringe (29-31G), swabs, sharps. |
| 2 | Clean stoppers | Wipe both stoppers. Air-dry 10-15 sec. |
| 3 | Draw 1mL BAC | Exactly 1 mL. Expel air bubbles. |
| 4 | Inject into vial | Aim at GLASS WALL. Slowly. Never onto powder. |
| 5 | Swirl & label | Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days. |




1mL Insulin Syringe | 29-31G | SubQ

**Formula:** Vol(mL)=Dose(mcg)/Conc(mcg/mL) | **Units:** Vol x 100

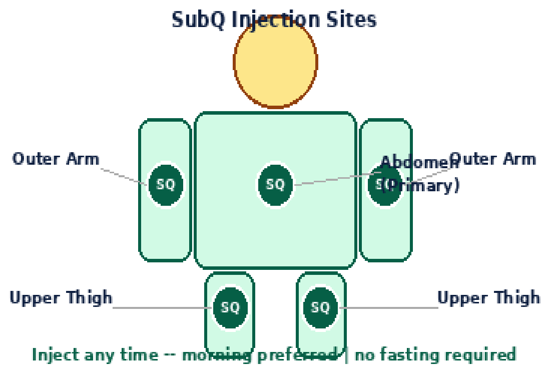
**10mg x 10** -- 1mL BAC -> 10.0mg/mL (10,000mcg/mL)

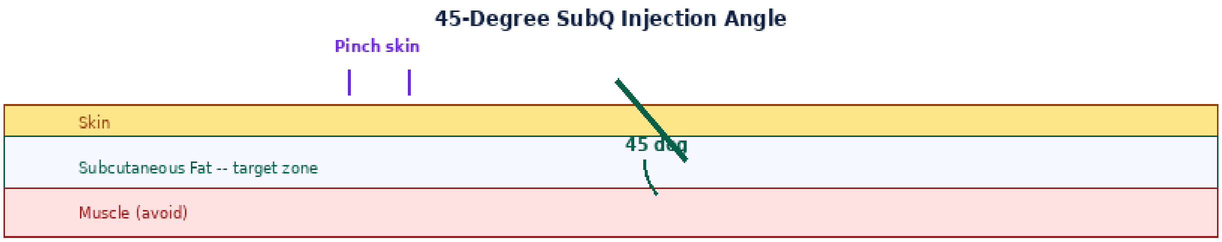
| Dose(mcg) | 1000mcg | 2000mcg | 3000mcg | 4000mcg | 5000mcg |
|---|---|---|---|---|---|
| **Vol(mL)** | 0.1000 | 0.2000 | 0.3000 | 0.4000 | 0.5000 |
| **Units** | 10.0U | 20.0U | 30.0U | 40.0U | 50.0U |
| **Doses/Vial** | 10 | 5 | 3 | 2 | 2 |

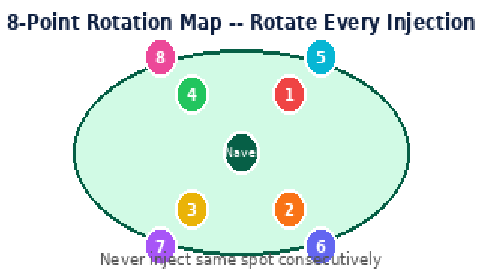
**40mg x 10** -- 2mL BAC -> 20.0mg/mL (20,000mcg/mL)

| Dose(mcg) | 1000mcg | 2000mcg | 3000mcg | 4000mcg | 5000mcg |
|---|---|---|---|---|---|
| **Vol(mL)** | 0.0500 | 0.1000 | 0.1500 | 0.2000 | 0.2500 |
| **Units** | 5.0U | 10.0U | 15.0U | 20.0U | 25.0U |
| **Doses/Vial** | 20 | 10 | 6 | 5 | 4 |

## INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES




| Step | Action | Detail |
|---|---|---|
| 1 | Wash & prepare | Wash hands. 29-31G insulin syringe. Draw exact dose volume. |
| 2 | Inject SubQ | Abdomen preferred. Pinch 2-3cm. 45 degrees. Inject slowly. |
| 3 | Rotate & dispose | Record site. 8-point rotation. Press 5 sec. Sharps container. |

| Protocol | Dose | Frequency | Duration | Notes |
|---|---|---|---|---|
| Standard | 1,000-2,000 mcg | 3-5x/week | 8-12wk | Morning injection preferred. |
| Research | 3,000-5,000 mcg | Daily | 8-12wk | Higher dose. Monitor response. |

| Side Effect | Likelihood | Management |
|---|---|---|
| Injection site reaction | Common | Rotate sites. Fresh needle each injection. |
| Fatigue (initial) | Occasional | Metabolic adaptation. Resolves within 1-2 weeks. |

**Contraindications: Cancer history, pregnancy, under 18.**

1. 10mg vial: 1 mL BAC Water yields 10 mg/mL (10,000 mcg/mL).
2. 40mg vial: 2 mL BAC Water yields 20 mg/mL (20,000 mcg/mL).
3. No fasting required -- MOTS-C acts independently of the GH/insulin axis.

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|---|
| 4. Inject 1x daily or 3-5x per week SubQ. Morning timing is preferred. |
| 5. Cycle 8-12 weeks on, followed by a mandatory 4-week off-cycle rest period. |
| 6. Store at 2-8 deg C refrigerated. Use reconstituted vial within 28 days. |
| 7. Rotate injection sites through all 8 abdomen points with every injection. |
| 8. Contraindicated in active cancer history -- AMPK and folate cycle modulation. |
| 9. Not for use during pregnancy, lactation, or in individuals under 18 years. |